A/C COMPRESSOR CLUTCH (Continued)

(9) Install the compressor shaft bolt. Tighten the bolt to 13 N·m (115 in. lbs.).

(10) Install the accessory drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - INSTALLATION).

(11) Reconnect the negative battery cable.

A/C CLUTCH RELAY

DESCRIPTION

The A/C clutch relay (Fig. 9) is a International Standards Organization (ISO) micro-relay. Relays conforming to the ISO specifications have common physical dimensions, current capacities, terminal functions and patterns. The ISO micro-relay terminal functions are the same as a conventional ISO relay. However, the ISO micro-relay terminal pattern (or footprint) is different, the current capacity is lower, and the physical dimensions are smaller than those of the conventional ISO relay.

The A/C clutch relay is located in the power distribution center (PDC) in the engine compartment.

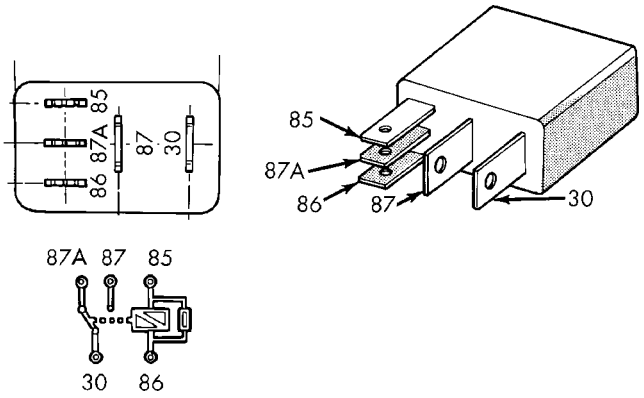

Fig. 9 A/C Compressor Clutch Relay

30 - COMMON FEED
85 - COIL GROUND
86 - COIL BATTERY
87 - NORMALLY OPEN
87A - NORMALLY CLOSED

OPERATION

The A/C clutch relay is an electromechanical switch that uses a low current input controlled by the powertrain control module (PCM) to control the high current output to the A/C clutch field coil. The movable, common feed relay contact is held against the fixed, normally closed relay contact by spring pressure. When the electromagnetic relay coil is energized, it draws the movable common feed relay contact away from the fixed, normally closed relay contact and, holds it against the fixed, normally open relay contact. This action allows high current to flow to the A/C clutch field coil.

When the relay coil is de-energized, spring pressure returns the movable relay contact back against the fixed, normally closed contact point. The resistor or diode is connected in parallel with the relay coil, and helps to dissipate voltage spikes and electromagnetic interference that can be generated as the electromagnetic field of the relay coil collapses.

The A/C clutch relay terminals are connected to the vehicle electrical system through a receptacle in the power distribution center (PDC). The inputs and outputs of the A/C clutch relay include:

- The common feed terminal (A10) receives battery current through a fuse in the PDC at all times.
- The coil ground terminal (A8) receives a ground input from the PCM through the A/C clutch relay control circuit only when the PCM electronically pulls the control circuit to ground.
- The coil battery terminal (A6) receives battery current through a fuse in the fuse block only when the ignition switch is in the Run or Start position.
- The normally open terminal (A9) provides battery current to the A/C clutch coil through the A/C clutch relay only when the A/C clutch relay coil is energized.
- The normally closed terminal (87A) is not connected to any circuit in this application, but provides a battery current output only when the A/C clutch relay coil is de-energized.

The A/C clutch relay cannot be repaired and, if faulty or damaged, it must be replaced. Refer to the appropriate wiring information for diagnosis and testing of the ISO-standard micro-relay and for complete HVAC wiring diagrams.